

Layer	Protocols in conventional networks	Protocols in constrained node networks
Data link	Ethernet, Wi-Fi	WirelessHart, ZigBee, BLE, Z-Wave, Thread, DASH7, HomePlug, G.9959, LTE-A, LoRaWAN, Weightless, Dect/ULE
Adaptation	NA	To encapsulate IPv6 datagrams, e.g., 6LoWPAN, 6TiSCH, 6Lo, IPv6 over BLE
Network	IPv4, IPv6, OSPF, RIP, etc	IPv6 was chosen rather than IPv4 to avoid address exhaustion issues
Transport	TCP, UDP	TCP is huge in size, so UDP is used in constrained node networks
Presentation	Data is represented using XML or JSON	Data is represented using CBOR (Concise Binary Object Representation)
Application	HTTP	CoAP (Constrained Application Protocol) is specifically designed for constrained nodes
Interoperability	NA	Applications use CoAP and its features in their own custom way. To support interoperability, the following standards are used -- Dotdot, OpenWeave, LWM2M

Table 3: Comparing protocols in conventional networks and constrained node networks

Technology	Link layer standard	Frequency	Max. range	No. of devices	Max. data rate	Native IP support	6LoWPAN support
BLE	Bluetooth SIG	2.4GHz	150m	32,000	1Mbps	No	Yes
Thread	IEEE 802.15.4	2.4GHz	30m	250-300	250Kbps	Yes	Yes
Z-Wave	Zensys	800-900MHz	30m	232	100Kbps	No	No
Zigbee	IEEE 802.15.4	900MHz - 2.4GHz	100m	65000	250Kbps	Yes	Yes

Table 4: Comparing constrained node technologies

The key points that make the Thread protocol the preferred technology are listed below:

- Thread is an open source implementation of an open specification.
- Thread has inherent IP support. Zigbee has recently got IP support as an enhancement.
- Thread and Zigbee support Dotdot, which is an application layer protocol for interoperability. The number of Zigbee devices in the world is significant. Since Thread supports Dotdot, it can also communicate with Zigbee devices in a seamless way.
- Thread has been supported by Google, ARM, Nordic, Cascoda, NXP, Particle, Qorvo, Qualcomm, Samsung, Silicon Labs, STMicroelectronics, Synopsys,

TI, Zephyr and Apple since 2018. This raises the hope that the Thread protocol will become the *de facto* technology for constrained nodes.

The first part in this series gives an introduction to IoT, constrained node networks, and how these networks are used in an IoT system. The

firmware requirements and the network protocols meant for constrained nodes have also been explored. The firmware comparison in Table 4 clearly shows that the Thread protocol has an edge over other technologies and is the recommended firmware for constrained nodes in IoT. **END** 🐧

References

- [1] NISTIR 8200: 'Interagency Report on Status of International Cybersecurity Standardization for the Internet of Things (IoT)'
- [2] NIST SP 800-183: 'Networks of Things'
- [3] RFC 7228 – Terminology for Constrained-Node Networks
- [4] White paper: 'Internet of Things Protocols and Standards' by Tara Salman under the guidance of Prof. Raj Jain
- [5] <https://openthread.io/>
- [6] [https://en.wikipedia.org/wiki/Thread_\(network_protocol\)](https://en.wikipedia.org/wiki/Thread_(network_protocol))

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